Week 1: Overview of LLMs & Training

Understand the fundamental building blocks of LLMs with tokenization, vectorization and attention.

- Tokenisation, Vectorization, Attention
- Pre-training and post-training
- LLM Evaluations
- End-to-end LLM lifecycle

tokens vectors attention LLMs

Week 2: Quantization and Fine-Tuning

Learn how LLMs are quantized for fast processing, and how to fine-tune models to meet specific business requirements.

- Post-training
- Quantization - FP16, attention optimizations
- Fine Tuning - LoRA/QLoRA
- Dataset prep → training → evaluation

LLM optimizations quantization LORA fine-tuning

Week 3: Retrieval Augmented Generation

Learn chunking strategies, data ingestion, reranking, indexing, vector databases, and other techniques for retrieval augmented generation.

- RAG: chunking strategies, data ingestion, reranker, indexer
- Vector Embeddings, Vector Databases
- Search Algorithms: ANN algorithms (HNSW, IVF)

RAG | Vector DB | Vector Search

Week 4: Hands-on RAG Implementation

An interactive project where students learn to code a RAG-based application and learn best practices for AI safety.

- Reranking strategies, Query rewriting, HyDE
- Input and output guardrails
- Safety: Prompt injection, Intent classification
- Coding Assignment: Build a RAG chatbot using API calls

RAG | safety | guardrails | coding

Week 5: AI Agents and Tool Calling

Learn what an Agent is, how they are different from plain LLMs, Tool Calling, ReAct pattern, and Agent Orchestration.

- LLM vs Agent vs Multiple Agents
- ReAct pattern
- Prompt Chaining, Orchestration, Routing
- Coding Assignment: Customer support agent

agents | reAct | orchestration | coding

Week 6: MCP, Context Engineering, Multi-Agent Systems

Code an AI Agent with MCP and memory, optimizing agentic flow.

- Context Engineering
- Memory in Agents
- Model Context Protocol
- Multi-Agents
- Coding Assignment: MCP with memory and optimising agentic flow

agentic memory | mcp | context engineering | memory systems

Week 7: Evals, AI Applications in Production

Learn how Evals are used in production AI applications, and best practices for AI development.

- Evals: How to avoid hallucinations with Evals
- LLM as a Judge
- Tradeoffs and design decisions
- Fine-tuning vs Prompting vs RAG
- Project: Build your own LLM Judge

Evals | LLM as Judge | hallucinations | AI in practice

Week 8: Agentic System Design

Learn how AI agents are scaled in distributed systems, and the system design of large-scale AI applications.

- Agents at scale
- MCP vs. API wrappers
- Design tradeoffs
- Best practices for agentic system design

Agents | MCP | System Design

Week 9: Image and Reasoning Models

Learn how multimodal models are trained with images and video, and the mechanism of diffusion-based models.

- Multimodal models
- CLIP
- Video Models
- CoT, RLHF

clip multimodal images rlhf

Week 10: Capstone Project

Create a production-grade AI project on a topic of your choice.

- Recap important concepts
- Problem selection
- Metrics for evaluation
- Feedback on completed projects

capstone project evaluation